

What are the global epidemiological trends in limb amputation rates, categorized by cause (diabetes, trauma, vascular disease) over the past decade?

Global traumatic limb amputation rates declined from 1990-2021 (0.26 to 0.24 per 100,000), but limited data exists on diabetes and vascular disease trends, with cause distribution varying by economic development level.

Abstract

Wei et al. (2025) report that traumatic limb amputations have grown in absolute numbers from 1990 to 2021, even as age-standardized incidence and prevalence rates declined. In 2021, 10 million new traumatic amputations and 445.2 million existing cases were recorded worldwide; the incidence rate dropped from 0.26 to 0.24 per 100,000 and is projected to decline further (from 138.5 to 113.0 per 100,000 by 2036). Regional differences emerge: South Asia shows the highest incidence and years lived with disability, while high-income regions (notably India and China) report higher prevalence. McDonald et al. (2021) present a 2017 snapshot estimating 57.7 million people living with traumatic limb amputations, with falls accounting for 36.2% of cases.

Findings regarding non-traumatic causes—diabetes and vascular disease—appear limited. Wei et al. (2025) note that in settings of higher economic development vascular disease contributes more to the overall amputation burden, whereas in developing regions trauma prevails.

Paper search

To answer your research question "What are the global epidemiological trends in limb amputation rates, categorized by cause (diabetes, trauma, vascular disease) over the past decade?", we looked over only the sources you uploaded.

3 sources were uploaded to Elicit for screening.

Screening

We screened in sources that met these criteria:

- **Amputation Data Reporting:** Does this study report limb amputation rates, incidence data, or epidemiological data on limb amputations?
- **Etiology Categorization:** Does this study categorize amputations by etiology, specifically including at least one of the following: diabetes, trauma, and/or vascular disease?
- **Study Timeframe:** Does this study include amputation data from 2015 onwards?
- **Temporal Trend Analysis:** Does this study report data that allows for trend analysis over time (e.g., data from multiple time points, years, or periods)?
- **Study Design Appropriateness:** Is this study one of the following designs: population-based study, registry study, hospital-based study, cross-sectional study, cohort study, systematic review, or meta-analysis?
- **Adequate Sample Size:** Is this study NOT a case report, case series with fewer than 10 patients, or individual case study?
- **Epidemiological Focus:** Does this study focus on epidemiological data rather than solely on surgical techniques, prosthetics, rehabilitation, or clinical outcomes without epidemiological context?
- **General Population Relevance:** Is this study NOT conducted exclusively in military or combat settings?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Geographic Scope:**

Extract the geographic coverage of the study including:

- Specific countries/regions included
- Whether coverage is global, regional, or national
- Any geographic limitations or exclusions mentioned
- Population representativeness claims

- **Time Period:**

Extract the temporal scope including:

- Start and end years of data collection/analysis
- Whether study covers the past decade (2014-2024)
- Any gaps in temporal coverage
- Baseline and endpoint years for trend analysis

- **Amputation Causes:**

Extract how amputations were categorized by cause including:

- Specific cause categories studied (diabetes, trauma, vascular disease, other)
- How causes were defined or classified
- Whether multiple causes were considered for same amputation
- Any cause-specific subcategories (e.g., types of trauma)
- Proportion or percentage attributed to each major cause

- **Rate Data:**

Extract all amputation rate/burden data including:

- Incidence rates (new cases per time period)
- Prevalence rates (existing cases)
- Units of measurement (per 100,000 population, absolute numbers, etc.)
- Age-standardized rates if provided
- Confidence intervals or uncertainty ranges
- Years lived with disability (YLDs) if reported

- **Trend Direction:**

Extract information about temporal trends including:

- Whether rates are increasing, decreasing, or stable over time
- Magnitude of change (percentage change, annual percentage change)
- Statistical significance of trends
- Any trend reversals or inflection points
- Projected future trends if provided

- **Data Sources:**

Extract methodology and data sources including:

- Primary data source (e.g., Global Burden of Disease, national registries, surveys)
- Data collection methods
- Quality assessment or limitations of data sources
- Whether data includes all amputation types or specific subsets
- Estimation methods used for missing data

- **Demographics:**

Extract demographic patterns including:

- Age distribution of amputations
- Sex/gender differences in rates or trends
- Socioeconomic factors (if analyzed)
- High-risk populations identified
- Any demographic groups showing different trend patterns

- **Key Findings:**

Extract the main results relevant to global amputation trends including:

- Overall summary of global trend direction
- Most significant findings about cause-specific trends
- Regional variations in trends
- Key drivers of observed changes
- Clinical or public health implications stated by authors

Results

Characteristics of Included Studies

Study	Study scope	Time period covered	Amputation causes examined	Key metrics reported	Full text retrieved
Wei et al., 2025	Global analysis using Global Burden of Disease (GBD) 2021 data; 204 countries/territories; projections	1990–2021 (with projections to 2036)	Traumatic (mechanical forces, falls); non-traumatic (diabetes, vascular disease)	Incidence, prevalence, age-standardized rates, years lived with disability (YLDs), projections, demographic and regional breakdowns	Yes

Study	Study scope	Time period covered	Amputation causes examined	Key metrics reported	Full text retrieved
McDonald et al., 2021	Global analysis using Global Burden of Disease (GBD) 2017 data	2017	Traumatic (with cause-specific breakdown, e.g., falls)	Prevalence, prevalence rate per 100,000, cause distribution (falls), implications for prosthetic need	No

Study scope and time period:

- Both studies conducted global analyses using Global Burden of Disease (GBD) data.
- Wei et al., 2025, covered 1990–2021 with projections to 2036 and included 204 countries/territories.
- McDonald et al., 2021, covered only 2017; we didn't find mention in the abstract of projections or number of countries for this study.

Amputation causes examined:

- Both studies examined traumatic causes.
- Only Wei et al., 2025, examined non-traumatic causes; we didn't find mention in the abstract of non-traumatic causes in McDonald et al., 2021.
- Both studies included falls as a traumatic cause.
- Only Wei et al., 2025, included mechanical forces, diabetes, and vascular disease.

Key metrics reported:

- Prevalence was reported in both studies.
- Incidence, age-standardized rates, years lived with disability (YLDs), projections, and demographic/regional breakdowns were reported in Wei et al., 2025; we didn't find mention in the abstract of these metrics in McDonald et al., 2021.
- Prevalence rate per 100,000, cause distribution, and implications for prosthetic need were reported in McDonald et al., 2021; we didn't find mention of these metrics in Wei et al., 2025.

Effects

Global Trends in Traumatic Amputation Rates

- Wei et al., 2025 :
 - Reports an increase in the absolute number of traumatic amputations globally from 1990 to 2021.
 - In 2021, there were 10 million new cases (7.7 million males, 3.2 million females) and 445.2 million existing cases (prevalence) (306.6 million males, 138.7 million females).
 - Age-standardized rates for incidence and prevalence declined significantly over this period (incidence: from 0.26 to 0.24 per 100,000; prevalence: from 0.24 to 0.22 per 100,000).
 - Years lived with disability (YLDs) in 2021 were 5.9 million (3.9 million males, 2.1 million females).

- Projections indicate a continued decline in global incidence (from 138.5 per 100,000 in 2021 to 113.0 per 100,000 in 2036) and prevalence (from 5333.4 to 4339.9 per 100,000).
- McDonald et al., 2021 :
 - Provides a global prevalence estimate for 2017: 57.7 million people living with traumatic limb amputation, with prevalence rates reported per 100,000 population.
 - We didn't find mention in the abstract of incidence or trend data.

Regional and Demographic Variations

- Wei et al., 2025 :
 - Identifies substantial regional and demographic variation.
 - South Asia has the highest incidence and years lived with disability (YLDs), while high-income countries have the highest prevalence.
 - India and China are among the countries with the highest burden.
 - Males, youth (highest incidence in 15–19 age group), and older adults (notably at age 75+) are high-risk groups.
 - The burden is higher in males, attributed to greater exposure to risk factors such as physical labor and violence.
 - Population growth and aging, especially in lower Socio-demographic Index (SDI) countries, are key drivers of increased burden.
- McDonald et al., 2021 :
 - States that the study aimed to analyze patterns by region and age, but we didn't find mention in the abstract of specific regional or demographic breakdowns.

Projections and Future Burden

- Wei et al., 2025 :
 - Projects a continued decline in age-standardized incidence and prevalence rates globally and in most regions over the next 15 years, despite ongoing increases in absolute numbers due to population growth.
 - The net drift for incidence is -0.55% per year.
 - Regional variations in trends are noted, with some areas experiencing increases and others decreases in age-standardized rates.
 - The association between Socio-demographic Index (SDI) and burden is not linear.
- McDonald et al., 2021 :
 - We didn't find mention in the abstract of projections or trend data.

Regional and Temporal Trends in Traumatic Amputation Rates

Geographic region / Study	Incidence rates (per 100,000)	Prevalence rates (per 100,000 / absolute)	Temporal trends (1990–2021 or 2017 snapshot)
Global (Wei et al., 2025)	10 million new cases in 2021; projected decline from 138.5 (2021) to 113.0 (2036)	445.2 million existing cases in 2021; projected decline from 5333.4 (2021) to 4339.9 (2036)	Age-standardized rates declined significantly 1990–2021; projected continued decline
South Asia (Wei et al., 2025)	Highest incidence and years lived with disability (YLDs)	No mention found	Regional variation; some areas with increasing, others with decreasing rates
High-income (Wei et al., 2025)	No mention found	Highest prevalence	No mention found
India, China (Wei et al., 2025)	Among highest burden	Among highest burden	No mention found
Global (McDonald et al., 2021)	No mention found	57.7 million people living with traumatic limb amputation in 2017	We didn't find mention in the abstract of trend data; 2017 snapshot only

Incidence Rates

- Quantitative incidence rates were available for global estimates in Wei et al., 2025.
- Qualitative statements about high incidence were found for two regions (South Asia and India/China, both from Wei et al., 2025).
- We didn't find mention of incidence rates for high-income regions (Wei et al., 2025) or in McDonald et al., 2021.

Prevalence Rates

- Quantitative prevalence rates were available for global estimates in both Wei et al., 2025, and McDonald et al., 2021.
- Qualitative statements about high prevalence were found for high-income regions and India/China (both from Wei et al., 2025).
- We didn't find mention of prevalence rates for South Asia (Wei et al., 2025).

Temporal Trends

- A reported decline in age-standardized rates was found in Wei et al., 2025 (global).
- Mixed or regionally variable trends were described for South Asia (Wei et al., 2025).
- We didn't find mention of temporal trend data for high-income regions, India/China (both from Wei et al., 2025), or in McDonald et al., 2021.

Geographic Coverage

- Both studies reported global data.
- Wei et al., 2025, also reported on South Asia, high-income regions, and India/China.

Overall, quantitative incidence and prevalence rates were only available for global estimates, while regional data were generally qualitative or not mentioned. Only Wei et al., 2025, provided temporal trend data for global rates, and described mixed regional trends; we didn't find mention of trend data for the other regions or for the 2017 global snapshot.

Cause-Specific Trends

- Wei et al., 2025 :
 - Identifies mechanical forces and falls as the leading causes of traumatic amputations, with falls being more injurious but less frequent.
 - The etiologic structure varies by economic development, with trauma more common in developing countries and vascular disease in developed countries.
 - We didn't find mention of detailed breakdowns for non-traumatic causes (such as diabetes or vascular disease).
 - McDonald et al., 2021 :
 - Reports that falls account for 36.2% of traumatic limb amputations globally, making them the leading traumatic cause.
 - We didn't find mention in the abstract of non-traumatic causes.
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Discussion

The available evidence from these two studies using Global Burden of Disease (GBD) data suggests a consistent picture of global trends in traumatic limb amputation:

- Both studies highlight trauma as a major cause, with falls as a leading contributor.
- Wei et al., 2025, provides a comprehensive analysis of trends, regional and demographic variation, and future projections.
- McDonald et al., 2021, provides a single-year prevalence estimate and cause distribution.
- The decline in age-standardized rates despite rising absolute numbers is a notable finding, suggesting possible improvements in prevention or care, but the ongoing impact of population growth remains a challenge.
- Evidence for non-traumatic causes (diabetes, vascular disease) is limited in the extracted data, and further clarification of trends in these categories is not possible from the available studies.

Limitations:

- Reliance on Global Burden of Disease (GBD) data.
- Potential data lag.
- Lack of detailed breakdowns for non-traumatic causes.
- Absence of trend and incidence data in McDonald et al., 2021, limits the ability to synthesize findings across all causes and time periods.

References

Baixing Wei, Jie Zhang, Yuanpei Cheng, Han Wu, Austin Porter, Soysal Baş, and Brooke Hergert. “Global, Regional and National Burden of Traumatic Amputations from 1990 to 2021: A Systematic Analysis of the Global Burden

of Disease Study 2021.” *Frontiers in Public Health*, 2025.

Cody L. McDonald, Sarah Westcott-Mccoy, Marcia R. Weaver, Juanita Haagsma, and Deborah Kartin. “Global Prevalence of Traumatic Non-Fatal Limb Amputation.” *Prosthetics and Orthotics International*, 2021.